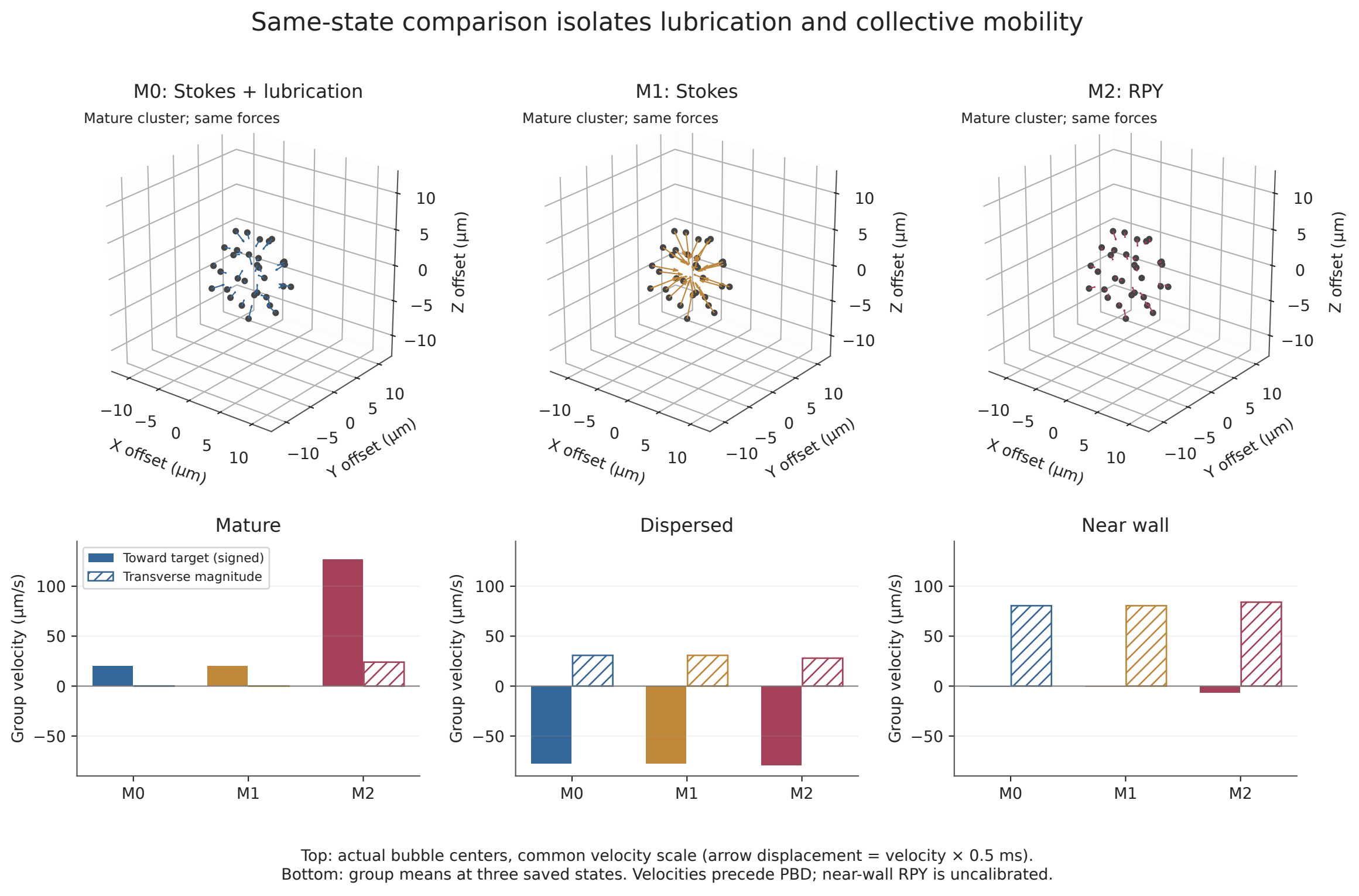
Same-state comparison isolates lubrication and collective mobility
M0: Stokes + lubrication
Mature cluster; same forces
M1: Stokes
Mature cluster; same forces
M2: RPY
Mature cluster; same forces
X offset (μm)
Y offset (μm)
Z offset (μm)
Mature
Dispersed
Near wall
Toward target (signed)
Transverse magnitude
Group velocity (μm/s)
M0
M1
M2
Top: actual bubble centers, common velocity scale (arrow displacement = velocity × 0.5 ms).
Bottom: group means at three saved states. Velocities precede PBD; near-wall RPY is uncalibrated.